PlatePlanner: Mobile App → IEEE Research Paper

A professor-pitchable plan to convert the PlatePlanner mobile project into an IEEE Access submission, anchored on the most under-researched slice of the work.

1. The Niche to Pitch (the "white space")

The existing recipe-recommendation and food-computing literature has explored single-recipe retrieval, ingredient substitution, and nutritional optimization in isolation — but **no prior system unifies pantry state, substitution feasibility, plan-level ingredient overlap, and food-waste minimization into a single multi-objective retrieval framework.** Pitch the paper around this gap, not the app.

> **Recommended working title:** *Sustainability-Aware, Pantry-State-Conditioned Multi-Objective Meal Planning — A Graph-Augmented Retrieval Framework with Plan-Level Evaluation Metrics.*

Why this lands with a professor:

- bullet It is **provably novel**: the related-work section in your existing draft already documents the gap.
- bullet It introduces a **reusable evaluation framework** (three plan-level metrics) — a citable methodological contribution.
- bullet It ties to **IEEE Access's societal-impact emphasis** via the food-waste angle.
- bullet You do **not** need to claim a new ML model. The contribution is the *problem formulation* + *metrics* + *system* — far easier to defend.

2. Structured Plan (9-Week Path to Submission)

Phase 1 — Re-scope the project as a research artifact (Week 1)

- bullet Reframe the mobile app as the *evaluation testbed*, not the contribution. The contribution is the retrieval framework + metrics.
- bullet Write a 1-page **research problem statement**: the gap, RQ1/RQ2/RQ3, and contributions C1/C2/C3.
- bullet Get explicit professor sign-off on (a) the niche, (b) the research questions, and (c) the venue **before** writing more.

Phase 2 — Lock the three contributions (Week 2)

- bullet **C1 — Pantry-aware retrieval objective:** formalize $S(q, r, P) = (1–w)·sim(q,r) + w·overlap(P, I_r)$; justify w=0.6 via grid search.
- bullet **C2 — Graph-augmented substitution:** formalize hybrid scoring score = α·direct + (1–α)·cooccur over a Neo4j ingredient graph; justify α=0.9.
- bullet **C3 — Plan-level multi-objective selection:** define PantryUtil, ShopBurden, WasteProxy plus a greedy-with-lookahead selector over 21 meal slots.
- bullet Each contribution must be **independently defensible** — that is what an ablation table buys you.

Phase 3 — Build the evaluation harness (Weeks 3–4)

- bullet Generate **300 reproducible scenarios** with fixed seeds, stratified by pantry size (small / medium / large) and dietary constraints.

- Implement **three baselines:** Semantic-Only (w=0), Overlap-Only (w=1), Hybrid-without-substitution (w=0.6).
- Build **three ablation variants:** –Substitution, –PlanLevel, –ShopBurden.
- Automate the metric pipeline so every code change re-runs the full results table.

Phase 4 — Run the experiments (Week 5)

- Produce Tables I–III (comparative, ablation, latency) — the skeleton is already in your draft; populate with real runs.
- Parameter-sensitivity sweep: $w \in \{0.3, 0.4, 0.5, 0.6, 0.7, 0.8\}$ and $\alpha \in \{0.5, 0.7, 0.9, 0.95\}$.
- Latency benchmark with Locust at 50 concurrent users to back the sub-100 ms p95 claim.

Phase 5 — Human evaluation (Weeks 5–6, in parallel)

- **20-participant within-subjects blinded preference study** (graduate students, 5 scenarios each).
- 5-point Likert across four dimensions: Relevance, Practicality, Substitution Quality, Overall Satisfaction.
- Get IRB / department review **now** — even a non-human-subjects-exempt letter strengthens the paper.

Phase 6 — Write the paper using your existing draft as scaffold (Weeks 7–8)

- Your draft already follows the right structure (Problem Formulation → Related Work → Architecture → Methodology → Setup → Results → Limitations → Future Work). Do **not** rewrite — fill it with real numbers.
- Tighten Related Work so each subsection ends with an explicit gap statement; reviewers reward this.
- Write a candid **Limitations** section (quantity-insensitivity, sense ambiguity, cuisine cross-contamination). Honest limitations *increase* acceptance odds.

Phase 7 — Pre-submission hardening (Week 9)

- Ship a **reproducibility package**: GitHub repo with scripts, seeds, scenario JSONs, and a `make reproduce` target.
- Run a **reviewer simulation**: one co-author reads the paper cold and writes a fake reviewer report; fix what they flag.
- Get professor / advisor sign-off as senior author before submission.

Phase 8 — Pick the right venue

Tier	Venue	Why
Primary	IEEE Access	Broad scope, fast turnaround, accepts strong system papers.
Backup A	ACM RecSys Workshop (Multi-stakeholder Recommendation)	Good fit for plan-level multi-objective framing.
Backup B	IEEE BigData — Food Computing track	Aligns with FoodKG / Recipe1M lineage.
Avoid	NeurIPS / ICML	Not a new-model paper; wrong audience.

3. Things to Deliberately *Not* Claim

Trimming overclaims is the single highest-leverage move for getting a systems paper approved. Stay in your lane:

- bullet **Do not** claim a new embedding model — you use `all-MiniLM-L6-v2`.
- bullet **Do not** claim a new LLM — RAG is plumbing here.
- bullet **Do not** claim production deployment outcomes — a 20-participant preference study is honest scope.
- bullet Frame the mobile app as an **implementation vehicle**, not a research contribution.

4. TL;DR (for the professor pitch)

- **Pitch the gap, not the app:** no existing system jointly optimizes semantic relevance + pantry state + substitution feasibility + food-waste minimization.
- **Three concrete contributions:** C1 pantry-aware retrieval objective, C2 hybrid graph + co-occurrence substitution, C3 plan-level multi-objective selector.
- **Three novel metrics** introduced as a reusable benchmark: PantryUtil, ShopBurden, WasteProxy.
- **Evaluation:** 300 reproducible scenarios + 3 baselines + 3 ablations + Locust latency benchmark + 20-person blinded user study.
- **Headline numbers already in your draft:** 70% fewer missing ingredients, 139% better pantry utilization, 66% lower shopping burden, sub-100 ms p95, 85% user preference.
- **Sustainability framing** (WasteProxy → food waste) aligns with IEEE Access's societal-impact emphasis — boosts acceptance odds.
- **Target venue:** IEEE Access primary; ACM RecSys workshop or IEEE BigData Food Computing as backup.
- **Approval lever:** get professor sign-off on niche + RQs + venue *before* writing more; show the ablation plan and IRB-exempt user study upfront.
- **Honest limitations** (quantity-insensitivity, sense ambiguity, cuisine cross-contamination) become Section IX future work — reviewers reward this.
- **9-week execution path:** re-scope → lock contributions → harness → experiments → user study (parallel) → fill the draft → hardening → submit.

5. One-Glance Timeline

Week	Milestone	Deliverable
1	Re-scope + professor pitch	1-page problem statement; sign-off
2	Lock C1 / C2 / C3	Formal definitions + hyperparameter plan
3-4	Evaluation harness	300 scenarios, 3 baselines, 3 ablations
5	Experiments + latency	Tables I-III populated with real numbers
5-6	User study (parallel)	20-person Likert results
7-8	Paper writing	Draft filled in with real results
9	Hardening + submit	Repro repo + reviewer simulation + submission

Prepared from the existing PlatePlanner IEEE Access draft. Anchored on the under-researched intersection of pantry-state retrieval, substitution feasibility, plan-level optimization, and food-waste minimization.